Paul Bergbusch

Data Science and AI Leader | Application Architect | Full-Stack Engineer

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Professional Experience

2019 – present **Senior Data Scientist & AI Developer | Triton Partners | Frankfurt, Germany**

- Designed, built and deployed customer-facing applications, including a RAG tool for company search, an investment research AI agent, and a leveraged buyout simulation app
- Developed full-stack front- and back-end scalable AI applications using Python (FastAPI, Flask, Dash, JWT), PySpark, Typescript (React.js, Next.js), large language models (Cohere, OpenAI, Perplexity, Gemini, LLaMa, Glean), vector databases (Elasticsearch), and cloud services (Azure DevOps, AWS Rekognition/SageMaker/Bedrock, Google Cloud Vision)
- Launched AI projects across European and UK portfolio companies, using computer vision, machine learning, and LLMs

2017 – 2019 **Head of Data Science | Deutsche Börse | Frankfurt, Germany**

- Led a data science team developing ML models for trade settlement prediction, NLP-driven classification, and customer support automation, resulting in new revenues and reduced costs

2012 – 2017 **Project Manager | Eurex Clearing | Frankfurt, Germany**

- Managed projects and engineered requirements for portfolio risk management solutions
- Developed mathematical models for cross-margining of OTC and exchange-traded interest rate derivatives, reducing margin requirements for customers

2008 – 2012 **Co-Founder & CTO | OTC Valuations | Vancouver, Canada**

- Built and operated a derivative valuation service *(sold to Tullett Prebon in 2010)*
- Led teams of analysts supporting customers and developers using JIRA, J2EE, C#, VBA, SQL, Numerix

2006 – 2008 **Quantitative Analyst | BlueCrest Capital Management | London, UK**

- Developed numerical models for pricing equity variance & volatility swaps and options in C++
- Developed trading strategies for equity-credit arbitrage

2004 – 2006 **Financial Engineer | FINCAD | Surrey, BC, Canada**

- Developed a Monte Carlo pricing engine and a calibration engine for interest rate derivatives in C++
- Built binomial and trinomial trees for pricing employee stock options and convertible bonds in VBA

2002 – 2004 **Software/Systems Engineer | MacDonald, Dettwiler and Associates | Richmond, BC, Canada**

- Designed the architecture, defined the software requirements, and prototyped a weather satellite data simulator (LIDAR)
- Developed and calibrated models of satellite motion, pointing, geo-location, and radiometry using Matlab and C++

2000 – 2001 **Postdoctoral Researcher | UBC/TRIUMF | Vancouver, BC, Canada**

- Found and measured the world's rarest decay of a subatomic particle (kaon decaying into a neutrino and an anti-neutrino)
- Analyzed terabytes of data to detect rare kaon decay patterns (1 in 10 billion) using PCA, DFA, and neural networks

Education & Certifications

- **Project Management Professional** (Project Management Institute, 2016)
- **Ph.D. Particle Physics** (University of British Columbia, November 2000) | 90.2%, A+
- **M.Sc. Nuclear Physics** (University of British Columbia, November 1995)
- **B.Sc. (Honors) Chemical Physics** (Simon Fraser University, November 1993) | GPA: 3.68
- **Associate degrees in piano/pipe organ performance** (Royal Conservatory of Toronto, 1986/1990)

Languages & Interests

- Native fluency in English; B1 level in German; basic Spanish
- Development of audio plugins (using JUCE, Claude Code) and mobile apps (using Kotlin, Swift)
- Passionate about electronic music, space science, western history, and eastern philosophy
- Avid skier, golfer, swimmer, squash and volleyball player